

# Lecture 3: Data Representation II — Addresses, Pointers & Arrays

Session 3 · Wed Sep 2, 2026

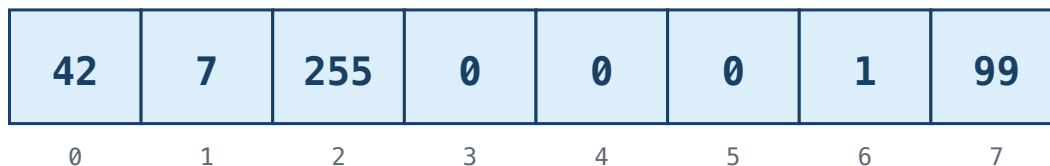
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## Memory Is a Row of Numbered Boxes

Last time we filled in one byte. Today: where the bytes *live*.

### Memory is a row of numbered boxes, one byte each



The small number under each box is its **address**. One box holds one byte.

Every byte your program can touch has a number — its **address**. Nothing more mysterious than house numbers along a street — though they run large, so we write them in hex: `0x7ffd4a08`, not `2147305992`.

#### Note

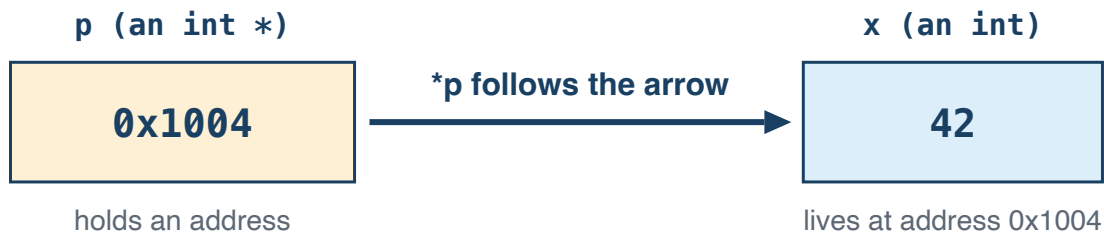
This row is your program's own **virtual address space**. Each running program gets one, and the operating system keeps them apart (*Lecture 9*). An optimizing compiler may also keep a value only in a register, where it has no address at all.

## Pointers Demystified

A **pointer** is a variable whose value is an address.

```
int x = 42;
int *p = &x;           // &x is "the address of x"
printf("%d\n", *p);    // 42 - *p is "the value at that address"
```

### A pointer is a box whose contents is an address



**&x** is the address of x. **\*p** is the value stored there.

### Three Things, Kept Straight

| Expression | Type  | Value            |
|------------|-------|------------------|
| x          | int   | 42               |
| &x         | int * | the address of x |
| *p         | int   | 42               |

Read the declaration `int *p` as “`*p` is an `int`” — which is exactly what the last row says.

#### Tip

`&` and `*` undo each other: `*&x` is `x`.

### Two Pointers, One Object

Nothing stops two pointers from designating the same object:

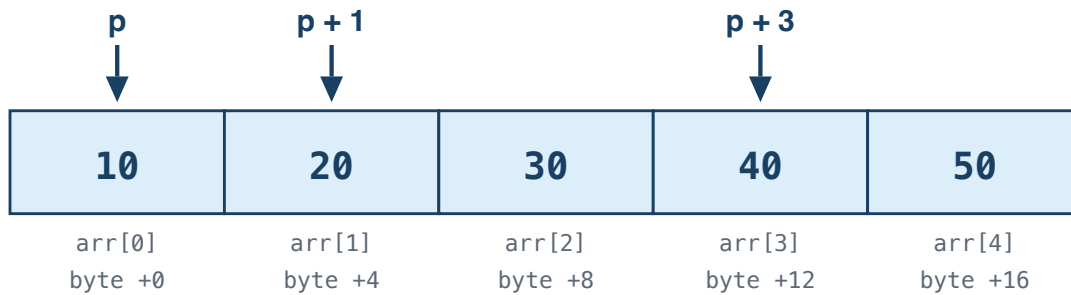
```
int x = 42;
int *p = &x;
int *q = &x;           // q points at the same int

*p = 99;
printf("%d\n", *q);    // 99 - q sees the change
printf("%d\n", x);     // 99 - and so does x
```

Two names, one box. This is called **aliasing**. It is why a function handed two pointers may not assume they point at different things — and exactly why `memmove` exists alongside `memcpy`, for the case where the two regions overlap.

## Pointer Arithmetic

**p + 1 moves one int, not one byte**



One int is 4 bytes here, so +1 steps 4 bytes. The compiler does the multiplying.

```
int arr[5] = {10, 20, 30, 40, 50};
int *p = arr;      // p points at arr[0]

*(p + 2) = 99;     // arr[2] becomes 99
```

**Rule:**  $p + i$  moves  $i * \text{sizeof}(*p)$  bytes — you count elements, the compiler counts bytes. Subtraction matches:  $\&\text{arr}[3] - \&\text{arr}[0]$  is 3, not 12. And  $a[i]$  is *defined* as  $*(a + i)$ , in the standard itself.

### Run It: the Addresses Step by Four

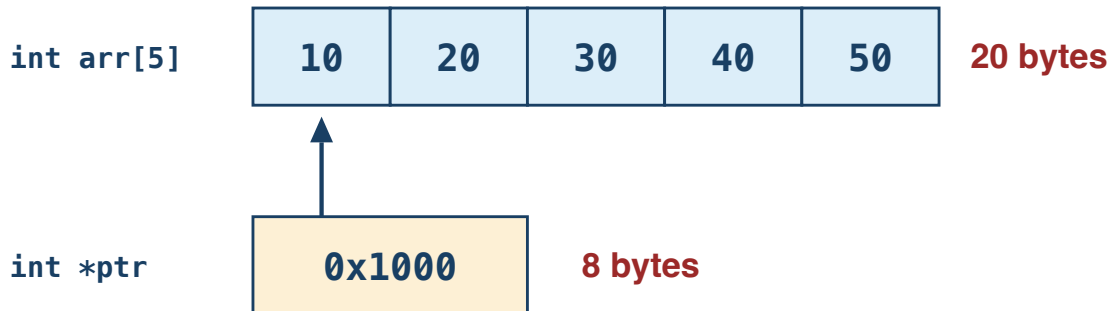
```
int arr[5] = {10, 20, 30, 40, 50};
for (int i = 0; i < 5; i++)
    printf("arr[%d] = %-3d  &arr[%d] = %p\n", i, arr[i], i, (void *)&arr[i]);
printf("&arr[3] - &arr[0] = %ld\n", (long)(&arr[3] - &arr[0]));
```

```
arr[0] = 10   &arr[0] = 0x16b106960
arr[1] = 20   &arr[1] = 0x16b106964
arr[2] = 30   &arr[2] = 0x16b106968
arr[3] = 40   &arr[3] = 0x16b10696c
arr[4] = 50   &arr[4] = 0x16b106970
&arr[3] - &arr[0] = 3
```

...960, 964, 968, 96c, 970 — four apart, every time. Your addresses will be different numbers; the spacing will not.

## An Array Is Not a Pointer

**An array is all the boxes. A pointer is one box.**



Both let you write `arr[2]`. Only one of them owns the storage.

```
int arr[5];           // five ints of storage
int *ptr = arr;       // one address, pointing into that storage

sizeof(arr);         // 20 on our teaching systems (5 × 4)
sizeof(ptr);         // 8 on our teaching systems

arr = ptr;            // ERROR: an array name is not assignable
```

## Arrays Decay to Pointers

In a parameter declaration, C **adjusts** `int a[]` to `int *a`. These two lines declare the same function:

```
void foo(int a[]);    // what you wrote
void foo(int *a);     // what the compiler sees
```

So the array-ness is gone by the time the function body runs:

```
void foo(int a[]) {
    sizeof(a);        // 8 here – a is a pointer
}
```

The length does not travel with the pointer. Pass it yourself:

```
void foo(int *a, size_t n);
```

## Strings Are Char Arrays

C has no string type. A “string” is a `char` array ending in a zero byte, `'\0'` — that terminator is how every library function knows where to stop.

```
#include <string.h>

char    arr[] = "hello"; // 6 bytes: h e l l o \0 – writable
const char *ptr = "hello"; // a pointer to a string literal

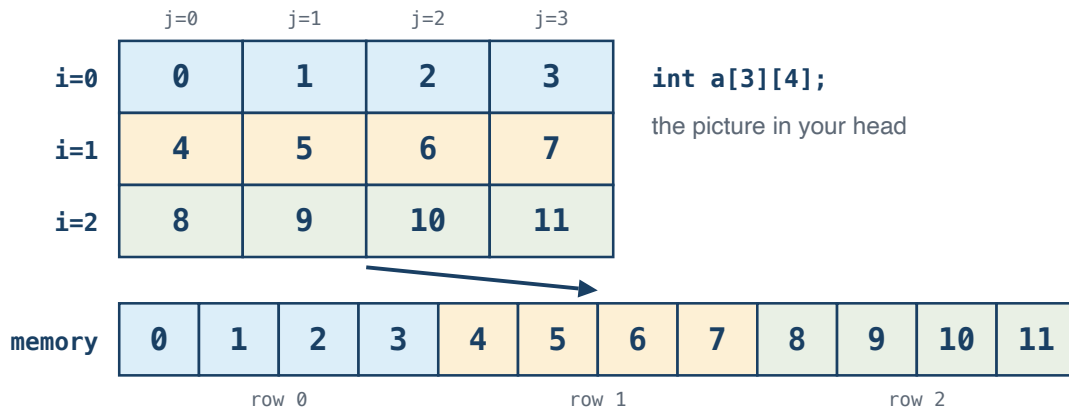
sizeof(arr); // 6 – the array, terminator included
strlen(arr); // 5 – characters before the terminator
sizeof(ptr); // 8 – just a pointer
arr[0] = 'H'; // fine
ptr[0] = 'H'; // won't compile – and that is the point
```

**Tip**

Say `const char *` for a string literal. Plain `char *` compiles too, but then writing through it is undefined behavior your compiler never mentions. `const` turns a crash you might not hit into an error you cannot miss.

**Two-Dimensional Arrays**

`int a[3][4]` is not a grid in memory. It is three rows of four, stored one row after the next — **row-major** order.

**A 2-D array is rows laid end to end**

**`a[i][j]` is element  $i*4 + j$**

Stepping `j` moves one slot. Stepping `i` jumps a whole row — four slots.

```
int a[3][4];
sizeof(a);      // 48 — the whole thing
sizeof(a[0]);   // 16 — one row of four ints
```

**Run It: Rows End to End**

```
int a[3][4];
for (int i = 0; i < 3; i++, printf(" <- row %d\n", i - 1))
    for (int j = 0; j < 4; j++) printf("%p ", (void *)&a[i][j]);
```

```
0x16fac6948 0x16fac694c 0x16fac6950 0x16fac6954 <- row 0
0x16fac6958 0x16fac695c 0x16fac6960 0x16fac6964 <- row 1
0x16fac6968 0x16fac696c 0x16fac6970 0x16fac6974 <- row 2
```

Read across, then down: `...954` ends row 0 and `...958` starts row 1 — the next four bytes. There is no gap between the rows, because there are no rows. It is one run of twelve ints.

**Why the Order Matters**

Walking a 2-D array along rows touches neighbouring bytes. Walking it down columns jumps a whole row every step:

```
for (i..) for (j..) sum += a[i][j]; // row-major: steps 4 bytes
for (j..) for (i..) sum += a[i][j]; // column-major: jumps 4*ncols bytes
```

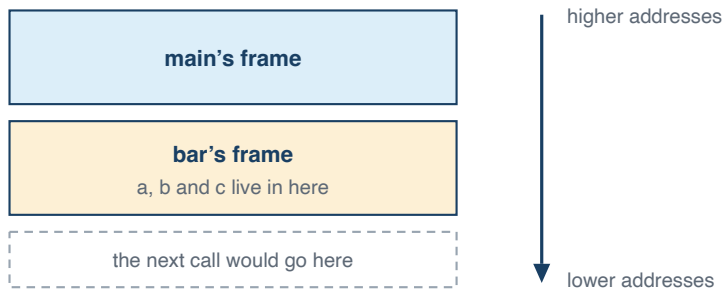
Same additions, same answer. In Lecture 1 the column-major version took **26 times longer** on a  $4096 \times 4096$  array. Now you can see why the two loops are not the same journey through memory — Lecture 16 explains what the hardware does about it.

## The Stack

Lecture 1 told you returning `&local` is broken and you took it on faith. This is where those locals live, and why the pointer outlives them.

```
void bar(void) {
    int a = 1, b = 2, c = 3;
}
```

### Every call gets its own frame



The stack grows downward. Where `a`, `b` and `c` sit *inside* the frame is the compiler's choice.

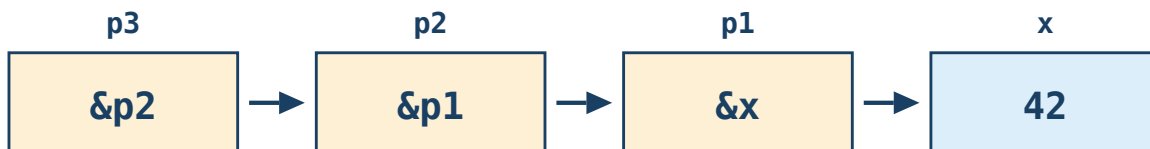
### Warning

The frame is real. Where `a`, `b` and `c` sit *inside* it is not — C leaves that to the compiler, which may even keep a value in a register with no address. Your machine gives the same layout every run, and that is the trap.

## Pointers to Pointers

```
int x = 42;
int *p1 = &x;    // pointer to int
int **p2 = &p1;  // pointer to pointer to int
int ***p3 = &p2; // and one more
```

### Each extra \* follows one more arrow



`*p3` is `p2` • `**p3` is `p1` • `***p3` is `42`

You have already used one: `main(int argc, char **argv)` receives a sequence of pointers, each pointing at one command-line argument.

## Null Pointers

NULL is the pointer value that means “points at nothing”.

```
int *p = NULL;
*p = 42;        // undefined behavior – on our systems, usually a crash
```

Check before you follow:

```
if (p != NULL) {  
    *p = 42;  
}
```

### Note

“Usually a crash” is the *good* case: the failure is loud and immediate. Undefined behavior is not required to crash at all.

### Practice

1. Given `int a[10]; int *p = a;` — why does `p + 1` refer to `a[1]` rather than to the next byte?
2. `p` is an `int *` holding `0x1000`. What address is in `p + 2`? What if `p` had been a `char *` instead?
3. Why does `sizeof` report 40 inside `main` but 8 inside a function you passed the array to?
4. `char *s = "hello"; s[0] = 'H';` compiles. What happens when you run it, and why does `char s[] = "hello"; s[0] = 'H';` behave differently?
5. Write a function taking `int **` that swaps which array two pointers refer to.

### Reading

- CS:APP §3.8 (Array Allocation and Access)
- The Descent to C — a careful pointer explainer